Rosen — §1.8: Proof Methods and Strategy
Selected Exercises

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Exercise 3

Prove that if x and y are real numbers, then $\max(x, y) + \min(x, y) = x + y$. [*Hint*: Use a proof by cases, with the two cases corresponding to $x \geq y$ and $x < y$, respectively.]

Exercise 7

Prove the triangle inequality, which states that if x and y are real numbers, then $|x|+|y| \geq |x+y|$ (where $|x|$ represents the absolute value of x, which equals x if $x \geq 0$ and equals $-x$ if $x < 0$).

Exercise 9

Prove that there are 100 consecutive positive integers that are not perfect squares. Is your proof constructive or nonconstructive?

Exercise 13

Prove or disprove that there is a rational number x and an irrational number y such that x^y is irrational.

Exercise 17

Suppose that a and b are odd integers with $a \neq b$. Show there is a unique integer c such that $|a - c| = |b - c|$.

Exercise 21

Prove that given a real number x there exist unique numbers n and ℓ such that $x = n - \ell$, n is an integer, and $0 \leq \ell < 1$.

Exercise 23

The *harmonic mean* of two real numbers x and y equals $2xy/(x+y)$. By computing the harmonic and geometric means of different pairs of positive real numbers, formulate a conjecture about their relative sizes and prove your conjecture.

Exercise 25

Write the numbers $1, 2, \ldots, 2n$ on a blackboard, where n is an odd integer. Pick any two of the numbers, j and k, write $|j - k|$ on the board and erase j and k. Continue this process until only one integer is written on the board. Prove that this integer must be odd.

Exercise 29

Prove that there is no positive integer n such that $n^2 + n^3 = 100$.

Exercise 33

Adapt the proof in Example 4 in Section 1.7 to prove that if $n = abc$, where a, b, and c are positive integers, then $a \leq \sqrt[3]{n}$, $b \leq \sqrt[3]{n}$, or $c \leq \sqrt[3]{n}$.

Exercise 35

Prove that between every two rational numbers there is an irrational number.

Exercise 37

Let $S = x_1y_1 + x_2y_2 + \cdots + x_ny_n$, where $x_1, x_2, \ldots, x_n$ and $y_1, y_2, \ldots, y_n$ are orderings of two different sequences of positive real numbers, each containing n elements.

(a) Show that S takes its maximum value over all orderings of the two sequences when both sequences are sorted (so that the elements in each sequence are in nondecreasing order).

(b) Show that S takes its minimum value over all orderings of the two sequences when one sequence is sorted into nondecreasing order and the other is sorted into nonincreasing order.

Exercise 45

Use a proof by exhaustion to show that a tiling using dominoes of a 4×4 checkerboard with opposite corners removed does not exist. [*Hint*: First show that you can assume that the squares in the upper left and lower right corners are removed. Number the squares of the original checkerboard from 1 to 16, starting in the first row, moving right in this row, then starting in the leftmost square in the second row and moving right, and so on. Remove squares 1 and 16. To begin the proof, note that square 2 is covered either by a domino laid horizontally, which covers squares 2 and 3, or vertically, which covers squares 2 and 6. Consider each of these cases separately, and work through all the subcases that arise.]

Exercise 47

Show that by removing two white squares and two black squares from an 8×8 checkerboard (colored as in the text) you can make it impossible to tile the remaining squares using dominoes.

Exercise 49

(a) Draw each of the five different tetrominoes, where a tetromino is a polyomino consisting of four squares.

(b) For each of the five different tetrominoes, prove or disprove that you can tile a standard checkerboard using these tetrominoes.